Project Report

1. Project Overview

This project analyzes the customer shopping behavior using the transactional data of 3900 purchases across various category of product. The aim is to uncover the insights from the shopping pattern, customer segment, age group, and subscription behavior to guide strategic business decision.

2. Dataset Summary

-Rows: 3900

-columns: 18

-key Features:

-Customer demographic (Age, Gender, Location, Subscription Status)

-Purchase details (Item purchased, Category, Purchase amount, Season, Size, Color)

- Shopping behavior (Discount applied, Promo Code used, Previous purchase,

Frequency of purchase, Review rating, Shipping type)

-Missing data 37 values in Review Rating column

2. Exploratory data analysis with python

We began with data preparation and cleaning in python:

- Data loading: Imported the data set using pandas
- Initial Exploration: Used df.info() to check the data structure and df.describe() for statistical summary

	customer_id	age	gender	item_purchased	category	purchase_amount	location	size	color	season	review_rating	subscription_status	shipping_type	discount_applied
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3900.000000	3900	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	2
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	No
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	2223
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750051	NaN	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.713590	NaN	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN

promo_code_used	previous_purchases	payment_method	frequency_of_purchases	age_group	purchase_frequency_date
3900	3900.000000	3900	3900	3900	3900.000000
2	NaN	6	7	4	NaN
No	NaN	PayPal	Every 3 Months	Young_Adult	NaN
2223	NaN	677	584	1028	NaN
NaN	25.351538	NaN	NaN	NaN	89.133077
NaN	14.447125	NaN	NaN	NaN	119.037566
NaN	1.000000	NaN	NaN	NaN	7.000000
NaN	13.000000	NaN	NaN	NaN	14.000000
NaN	25.000000	NaN	NaN	NaN	30.000000
NaN	38.000000	NaN	NaN	NaN	90.000000
NaN	50.000000	NaN	NaN	NaN	365.000000

- Handling the missing values: Checked for the null values in each column and imputed the missing value in Review Rating with the median rating of each product category.
- Column Standardization: Renamed each column in **Snake case** for better readability and documentation.
- Feature Engineering:
    - Created **Age_group** column by binning the customer ages.
    - Created **Freguency_of _purchase** from purchase data.
- Data Redundancy check: Verified if Discount_applied and Promo_code_used were redundant, dropped Promo_code_used
- Database integration: Connect python script to My SQL and loaded the cleaned data frame into database for SQL analysis

4. Data Analysis using SQL (Business Transactions)

We performed structured analysis in My SQL to answer key business questions:

1. Revenue by Gender – Compared total revenue generated by male vs female

gender	revenue
Male	157890
Female	75191

2. High-Spending Discount Users – Identified customers who used discount but still spend above the average purchase amount.

customer_id	purchase_amount
2	64
3	73
4	90
7	85
9	97
12	68
13	72
16	81
20	90
22	62
24	88
29	94
32	79
33	67
35	91
37	69

3. Top five products by ratings – Found products with the highest review ratings

item_purchased	average product rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.8
Skirt	3.78

4. Subscribers' vs non-subscribers - Compared average spend total revenue across subscribers' status

	subscription_status	count(customer_id)	Average_spend	Total_Revenue
▶	Yes	1053	59.49	62645
	No	2847	59.87	170436

5. Discount-Dependent Products – Identified 5 products with the highest percentage of discounted purchases

item_purchased text	discount_rate numeric
Hat	50.00
Sneakers	49.66
Coat	49.07
Sweater	48.17
Pants	47.37

6. Customer-Segmentation – Classified customer into New , Returning and Loyal segment based on purchase history

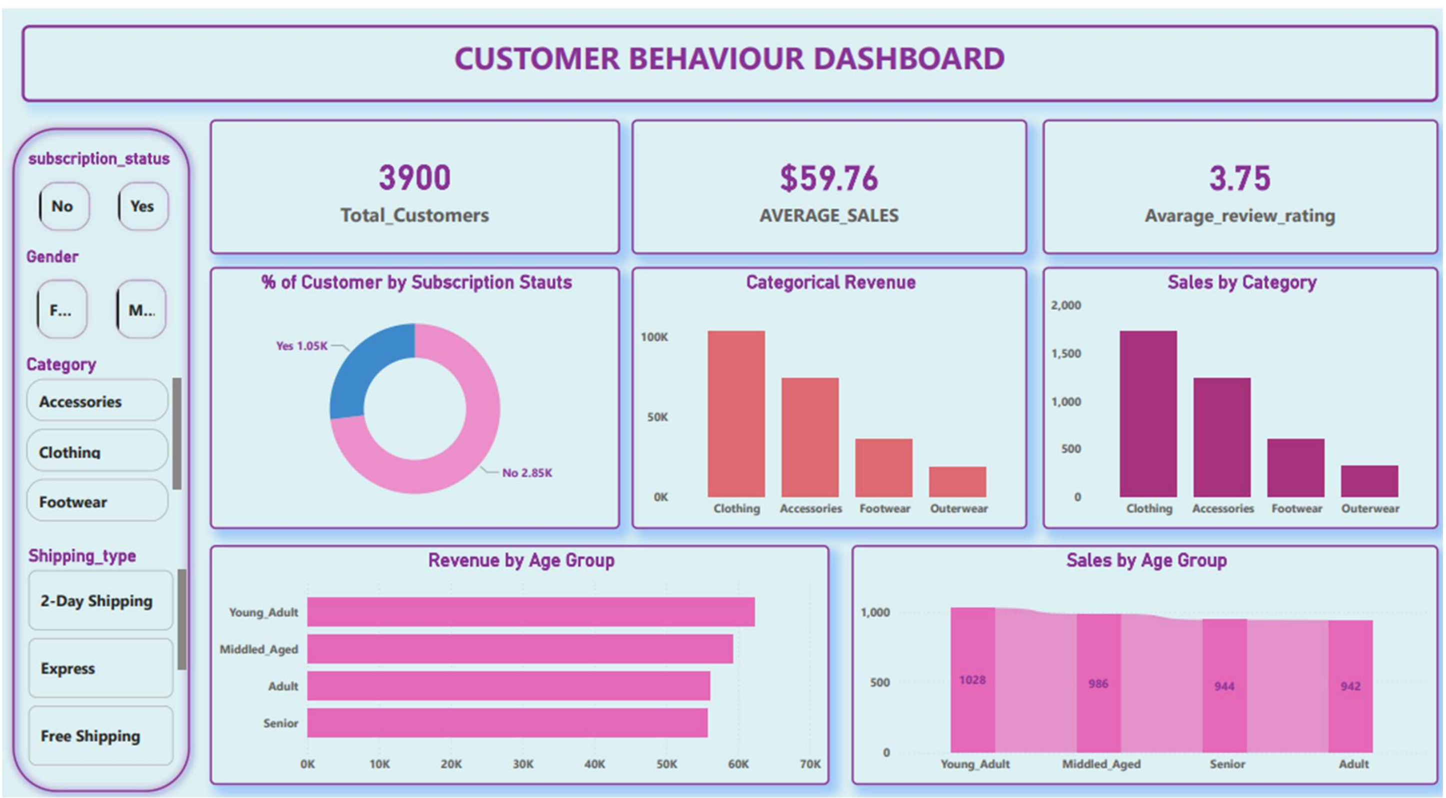
	customer_segment	Number Of Customer
▶	Returning	1478
	Loyal	2339
	New	83

7. Repeated Buyers & Subscription – checked weather the customer with >5 purchases are more likely to subscribe

subscription_status text	repeat_buyers bigint
No	2518
Yes	958

5. Dashboard In Power Bi

Finally, we prepared a dashboard in power bi to evaluate the insights.



6. Business Recommendation

1. Boost subscription – Promote exclusive benefits for Subscribers.
2. Customer Loyalty Programs – Reward repeated buyers to move them to move them them into 'Loyal' customers.

3.Review Discount policy – Balance sales boost with marginal control.

4.Product Positioning – highlight the best-selling and top-rated product.